Theorem: $e(G) \geq 3$ and $\alpha(G) \leq \kappa(G) \Rightarrow G$ has a Hamiltonian cycle.

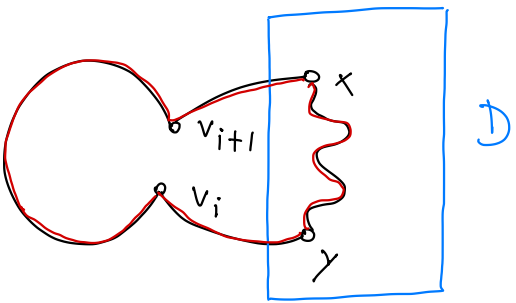
proof: Let C be the longest cycle in G. Set $\ell := v(C)$ and

$$V(C) = \{ v_i : i \in \mathbb{Z}_\ell \} \quad \text{and} \quad E(C) = \{ v_i v_{i+1} : i \in \mathbb{Z}_\ell \}$$

Suppose for contradiction that $\ell \neq v(G)$.

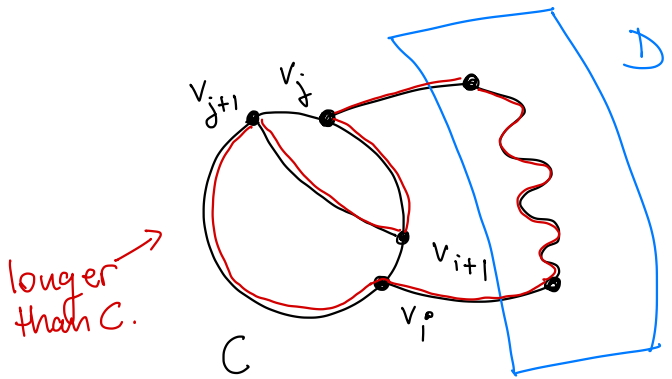
Let D be a component of $G - C$ and set $I := \{ i : v_i \in N_G(D) \}$

Obs 1: $\{i, i+1\} \not\subseteq I \ \forall i \in \mathbb{Z}_\ell$



This contradicts the maximality of C.

Obs 2: $\{ v_{i+1} : i \in I \}$ is ind. set.



Oth, contradiction again!

- Obs 1 + 2 $\Rightarrow$ $A := \{v_{i+1} : i \in I\} \cup \{v\}$ is an ind. set $\forall\, v \in D$.
- $B := \{v_i : i \in I\}$ $\Rightarrow$ B is a separator, and hence $|B| \geq \kappa(G)$.
- $|A| = |B| + 1 \leq \alpha(G) \leq \kappa(G)$, contradiction $\blacksquare$

<u>Defn</u>: A directed graph (or digraph) is a pair (V, E) where

$$E \subseteq \{(u,v) : u, v \in V,\ u \neq v\}.$$

<u>Capacity function</u>: $c : E(G) \to \mathbb{R}_{>0}$, where G is a digraph.

<u>Defn</u>: Let G be a digraph, $s, t \in V(G)$, $s \neq t$, and $c : E(G) \to \mathbb{R}_{>0}$

We say that a function $f : V(G) \times V(G) \to \mathbb{R}$ is an <u>(c, s, t)-flow</u> in G if

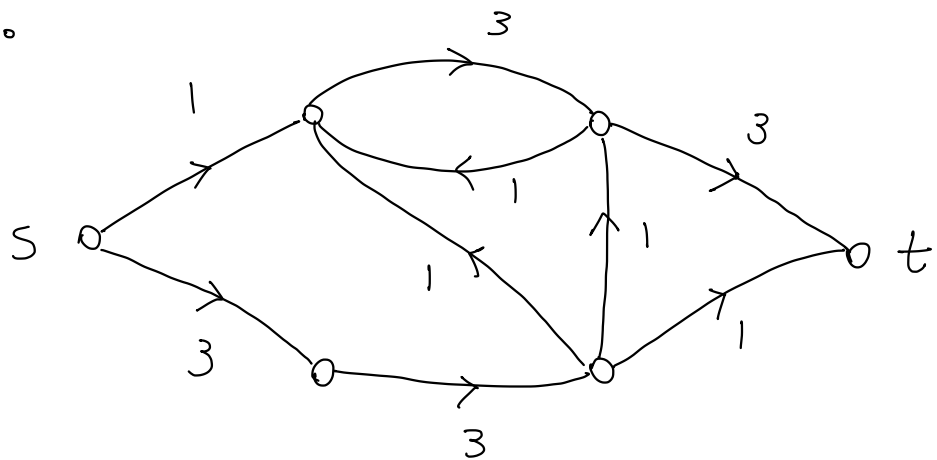
source sink

① $0 \leq f(u,v) \leq c(u,v)\ \forall\, (u,v) \in E(G)$.

② $f(u,v) = 0\ \forall\, (u,v) \notin E(G)$

② $\sum_{u \in V(G)} f(v,u) = \sum_{u \in V(G)} f(u,v) \quad \forall\, v \in V(G) \setminus \{s,t\}$ Kirchhoff's current law.

<u>Example</u>:



scheme of a flow in a graph (capacities omitted). (we omit $f(u,v)=0$)

<u>Defn</u>: The value of a (c,s,t)-flow f in G is

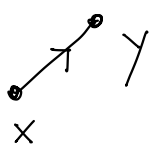
$$|f| := \sum_{u \in V(G)} f(s,u) - \sum_{u \in V(G)} f(u,s)$$

<u>Lemma</u>: Let f be a (c,s,t)-flow in a digraph G. Then,

$$|f| = \sum_{u \in V(G)} f(u,t) - f(t,u)$$

<u>proof</u>:

$$|f| = \sum_{x \in V(G) \setminus \{t\}} \left(\sum_{u \in V(G)} f(x,u) - f(u,x) \right) = \sum_{u \in V(G)} f(u,t) - f(t,u).$$



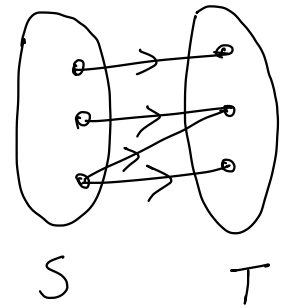
$$f(x,y) - f(x,y) = 0$$

(contribution of edges in $G[V(G) \setminus \{t\}]$)

$\blacksquare$

Defn: A s-t cut of G is a partition $V(G) = S \cup T$ with $s \in S$ and $t \in T$

For $c : E(G) \to \mathbb{R}_{>0}$, we write $c(S,T) := \sum_{\substack{(u,v) \in E(G): \\ u \in S,\, v \in T.}} c(u,v)$.



We refer to $c(S,T)$ as the capacity of the cut (S,T).

Lemma: Let f be a (c,s,t)-flow in a digraph G and let (S,T) be a cut in G. Then, $c(S,T) \geq |f|$.

proof:

$$0 = \sum_{u \in \underbrace{\cancel{S \setminus \{s\}}}_{S}} \left(\sum_{v \in V(G)} f(u,v) - f(v,u) \right) - |f| + \cancel{|f|} \leq c(S,T) - |f|.$$

$$\sum_{u \in S} \left(\sum_{v \in V(G)} f(u,v) - f(v,u) \right) = \sum_{\substack{u \in S, \\ v \in T}} f(u,v) - f(v,u) \leq c(S,T).$$

$\blacksquare$

<u>Thm</u> (Ford, Fulkerson) Let G be a digraph, $s,t \in V(G)$, $s \neq t$ and $c : E(G) \to \mathbb{R}_{\geq 0}$.

Then,

$$\max\{ |f| : f \text{ is } (c,s,t)\text{-flow} \} \geq \min\{ c(S,T) : (S,T) \text{ is an } s\text{-}t \text{ cut} \}$$

Goal

A flow f st $|f|$ is maximum is called a maximum flow

An s-t cut such that $c(S,T)$ is minimum is called a minimum cut.

This is also called the max-flow min-cut thm.

Q/ Why a max. flow exists?

<u>Real analysis</u>: Let $D \subseteq \mathbb{R}^m$ be a compact and $f: D \to \mathbb{R}$ be a continuous function.

Then $\exists\, x$ st $f(x) = \max_{y \in R} f(y)$.

Consider the space $\mathbb{R}^{V(G) \times V(G)}$, where coordinates are indexed by $(x,y) \in V(G) \times V(G)$.

For $u \in V(G)$, let $R_u := \left\{ \vec{x} \in \mathbb{R}^{V(G) \times V(G)} : \sum_{v \in V(G)} x_{uv} = \sum_{v \in V(G)} x_{vu} \text{ and } x_{ab} = 0 \ \forall ab \notin E(G) \right\}$

closed set.

Let $C = \left\{ \vec{x} \in \mathbb{R}^{V(G) \times V(G)} : x_{uv} \in [0, c(u,v)] \ \forall uv \in E(G) \text{ and } x_{uv} = 0 \text{ oth.} \right\}$

closed and bounded.

$\Rightarrow D := C \cap \bigcap_{v \in V(G)} R_v$ is compact and every $\vec{x} \in D$ gives a flow.

Define $g : D \to \mathbb{R}$ as $g(\vec{x}) = \sum_{v \in V(G)} x_{sv} - x_{vs}$ ← continuous function.

<u>proof of max-flow min-cut</u>: Let $f : V(G) \times V(G) \to \mathbb{R}$ be a maximum (c,s,f)-flow.

<u>Goal</u>: to construct a cut (S,T) such that $c(S,T) = |f|$.

Let H with $V(H) = V(G)$ and $E(H) = \{(u,v) : f(u,v) < c(u,v) \text{ or } f(v,u) > 0\}$

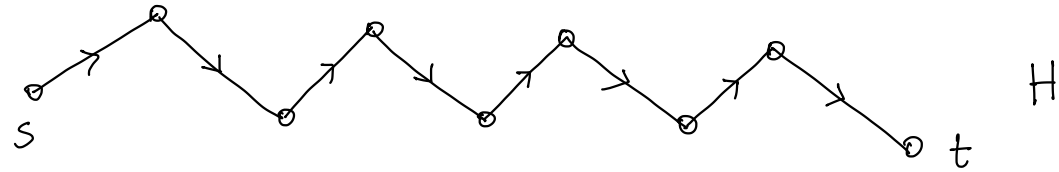
In other words, $(u,v) \in E(H)$ if we can increase $f(u,v) - f(v,u)$ without violating $0 \leq f(x,y) \leq c(x,y)$.

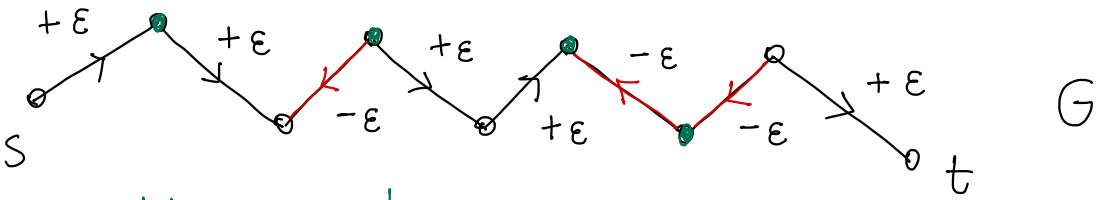
$S := \{v \in V(H) : \exists\, s \to v \text{ path in } H\}$

<u>Claim</u>: $t \notin S$.

<u>proof</u>: Suppose not. Let P an $s \to t$ path in H.

$\varepsilon := \min\{c(u,v) - f(u,v) : (u,v) \in E(P),\ f(u,v) < c(u,v)\} \wedge \min\{f(v,u) : (u,v) \in E(P),\ f(v,u) > 0\}$

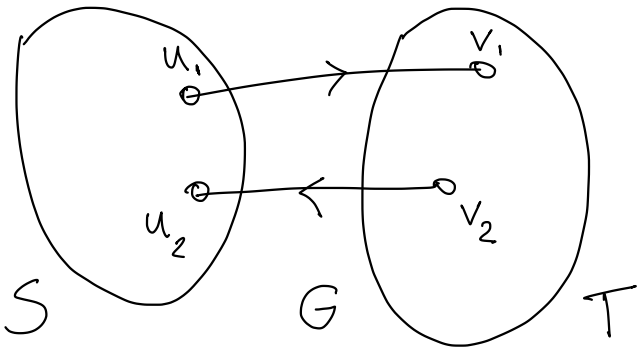




All configurations a vtx can see.

Set $T := V(G) \setminus S$. Claim $\Rightarrow$ (S,T) is a cut.

Defn of S $\Rightarrow$



- $f(u_1,v_1) = c(u_1,v_1)$
- $f(u_2,v_2) = 0$.

$$\sum_{\substack{u \in S \cancel{\setminus \{s\}} \; S \\ v \in V(G)}} f(u,v) - f(v,u) \; \cancel{+|f|} - |f| = 0$$

$$|f| = \sum_{\substack{u \in S \\ v \in V(G)}} f(u,v) - f(v,u) = \sum_{\substack{u \in S \\ v \in T}} f(u,v) - f(v,u)$$

$$= \sum_{\substack{u \in S \\ v \in T}} c(u,v) = C(S,T)$$

$\blacksquare$